1. Remove grease by agitating the head in a bath of commercial heavy duty alkaline, as sold by trade chemical suppliers.
2. Wash in hot water or steam clean to remove alkaline.
3. Remove scale by agitating the head in a bath of inhibited commercial pickling acid as sold by trade chemical suppliers. A 50-50 solution of Oakite 32 and water has been found satisfactory. After head is cleaned of grease and scale, it must be rinsed and the cleaner acid neutralized. Carefully follow directions of chemical manufacturer on cleaning, rinsing, and neutralizing.
4. Wash head in a bath of clean water or steam clean.
5. After head is inspected for cracks, dip in rust preventive to prevent rusting. Castings free of grease and oil will rust immediately when exposed to atmosphere.

INSPECTION

The information described under the following headings pertain to the cylinder head after it has been removed from the engine, disassembled and thoroughly cleaned.

WARPAGE (FLATNESS)

The cylinder head must be flat within certain tolerances longitudinally and transversely in order to ensure proper contact with the cylinder head gaskets and cylinder block to obtain the desired sealing effect. In continued service, cylinder heads may assume a contour to match that of the cylinder block and gaskets over a prolonged period of operation. This is a normal reaction. However, if the cylinder heads are allowed to become overheated because of coolant loss, high temperature causes stresses to occur in the castings which affects the flatness of the cylinder head. The maximum allowable variation in cylinder head flatness is shown in the following table:

Engine	Longitudinal	Transverse
4-71	.008″	.004″
6-71	.010″	.004″

When determining or measuring the flatness of the cylinder head fire deck, the use of a heavy, accurate straight edge and accurate feeler gauges should be employed. Measurements should be taken longitudinally at each bridge between cylinders and on both right and left sides of the cylinder head (fig. 73).

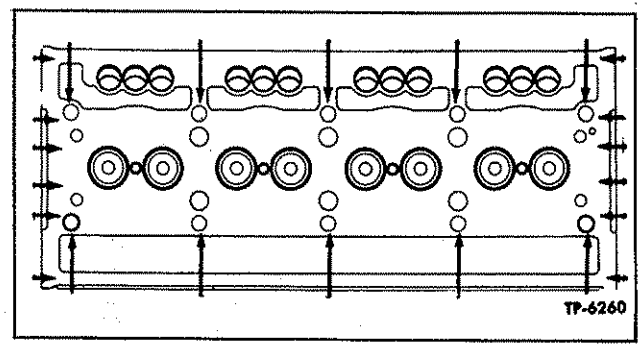


Figure 73—Check Cylinder Head Flatness at Points Indicated

If the flatness is not within the dimensions given in the table, then the fire deck of the cylinder head should be refaced. A limit of .015″ to .020″ is the maximum amount of stock that should be removed. Critical dimensions such as protrusion of valve seats, valves, injector hole tubes, and injector spray tips should be observed. In the event a cylinder head is not flat within its re-use limits then head should be replaced.

LEAKS

Following the inspection of the cylinder head for flatness as described above, the next step that should be performed is checking the cylinder head for leaks. Install dummy injectors, tap lightly, and hold in place with injector crab and nut tightened to recommended torque. Dummy injectors may be made up with old injector nuts and bodies—the injector spray tip is not necessary. Remove pipe plugs at both ends of the cylinder head oil gallery.

Seal all water openings in the cylinder head with suitable gasket material and clamping fixture equipment; however, provision should be made to introduce air into the water passageway cavities of the cylinder head as it is being tested.

The cylinder head assembly should then be submerged into water heated to a temperature of 180° to 200° for about fifteen to twenty minutes to thoroughly heat the cylinder head. Any leaks in the cylinder head assembly will be detected by allowing air to enter the cylinder head water passageways until a pressure of 80-100 lbs. (maximum) has been attained and noting any air bubbles in the water. Some of the places to inspect for leaks are as follows: Bottom and top of the injector tubes; oil gallery; cylinder head stud holes; exhaust ports; top, bottom, and sides of cylinder head proper.